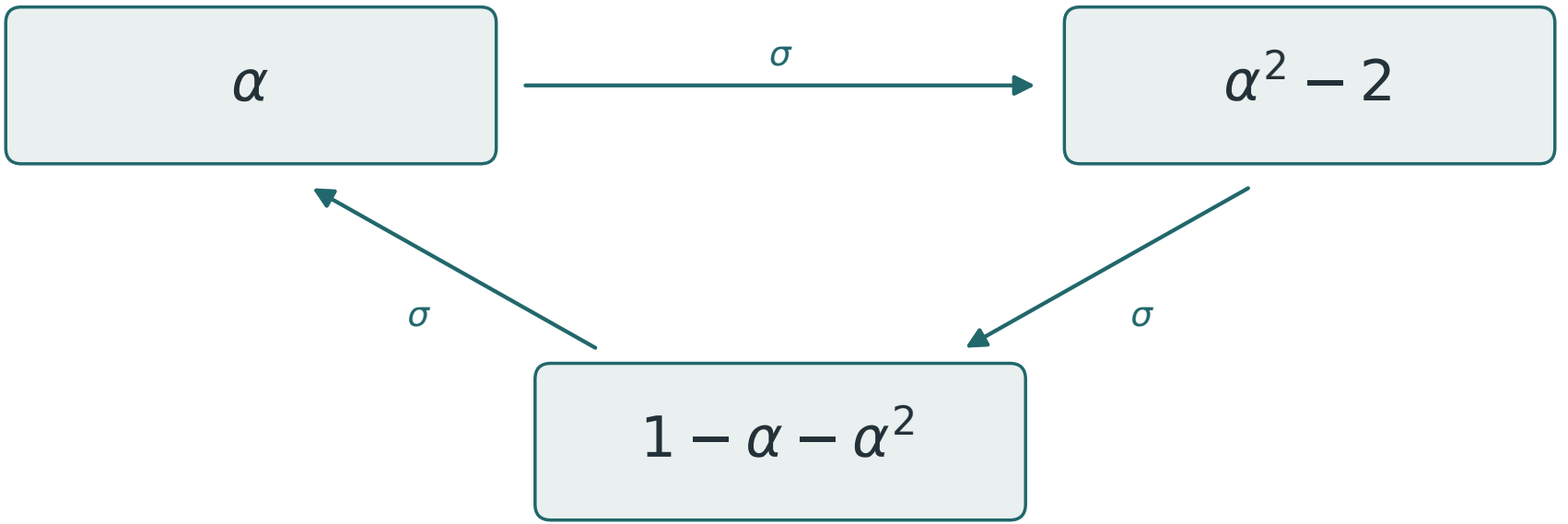
A cubic whose three roots form one cycle

WRITTEN PROOF / an irreducible cubic with square discriminant

$$f(x) = x^3 + x^2 - 2x - 1, \qquad \Delta = 49$$



$$\sigma^3 = 1, \qquad L = \mathbb{Q}(\alpha), \qquad \mathrm{Gal}(L/\mathbb{Q}) \cong C_3$$